CFM 301: Financial Data Analytics

Course Projects — Instructions

Over the term you will complete a series of hands-on projects that build your ability to source, clean, and analyze financial data, and to turn that analysis into an investment decision. Each project builds on the tools covered in the corresponding module. Beyond the objective stated for each project, the choice of data sample, asset universe, and implementation details is left to you — document and justify your choices.

Purpose of the Projects

The main objective of each project is to assess how well your group has grasped the class material and how correctly you apply it. Each project lists the techniques from class you are expected to apply (e.g., constructing returns correctly, cleaning the data, testing how sensitive your results are to your assumptions), and grading follows that list. If a technique does not apply to your setting, explain why in the report; a technique skipped without explanation is treated as missing.

Economic Question, Motivation & Audience

Crucially, every project needs to have a specific economic goal or research question. It is everyone's job in the group to come up with and refine this question.

Formulating a clear, well-motivated economic question is what sets **CFM** apart from more technical programs such as **FARM** (Mathematics/Financial Analysis and Risk Management). While technical rigor is essential, CFM projects must bridge quantitative techniques with real-world financial decision-making and economic insight.

Intended audience: State who your analysis is for and who would be interested in funding it (e.g., a government agency, the general public, mutual fund or pension fund managers). Your audience should shape how you frame the question, which assumptions are reasonable, and how you word your conclusion.

Example: In a recent discussion on capital allocation (https://www.youtube.com/watch?v=KCt9a9AnNfk), Mark Carney mentioned that global capital is under-weighted in the Canadian market relative to its benchmark. Is that claim empirically true? A project could conduct a quantitative portfolio optimization exercise specifically designed to test this hypothesis—evaluating whether an under-allocation to Canadian assets is justified by risk-return trade-offs or represents an unexploited market inefficiency. Possible audiences: federal policymakers or Canadian pension fund managers.

Reproducibility

The instructor must be able to run your code on the files you submit and get the numbers in your report. Reproducibility is a separate item in every project rubric. Before you submit:

- Submit the data files your code needs, but include only the data you actually use (e.g., the assets, variables, and sample period in your analysis), not the whole dataset you downloaded. Your code should run on these files without a WRDS login, an internet connection, or any manual step (no copy-pasting between files, no editing in Excel).
- Use relative file paths (e.g., data/crsp_monthly.csv), never paths on your own computer (e.g., C:\Users\...\Desktop).
- Include a short README listing the language and version, required packages (with versions), and which files to run in what order.
- Make sure your code reproduces every table and figure in the report from a clean start (for notebooks, use "Restart & Run All").
- Have a group member who did not write the code run your submission from a new folder, start to finish.

If your results cannot be reproduced from your submission, the reproducibility points are lost, and results that cannot be verified may not receive credit in other rubric categories.

Groups, Submissions & In-Class Presentations

These are group projects. You will work in groups of 3–5 students; there are 11 groups in total. Group membership stays the same for all projects.

Submission: Upload your report, code, data files, and README to the project's LEARN Dropbox folder no later than 2 days before the presentation date. Upload your presentation slides to the separate LEARN Dropbox folder for slides no later than 1 hour before the presentation session begins.

Presentation: Your group will deliver a 5–6 minute in-class presentation of your own project: your economic question and audience, your data and methodology, your results, and your economic conclusion. Across the 6 projects over the term, every student in the group must present in class at least once.

Attendance and individual questions: After your presentation, the instructor will ask every member of your group at least one question. All group members must therefore attend their group's presentation session. A member who is absent cannot be asked a question and will receive a lower presentation grade for that project.

Peer review: For each project, your group is paired with a different group. Read your paired group's report after the report deadline, and during their presentation ask at least one substantive question or make one substantive comment on their methodology, assumptions, or conclusions.

Each project is worth 15 points total: 12 points for your own project (analysis, report, and code) and 3 points for your in-class presentation and peer review — see the grading rubric for each project below.

Project 4: Reproducing the Credit Spread / Liquidity Figure

Objective

Reproduce the figure on slide 59 of Ch. 9 (Bonds) — "Baa-Tsy Yield Spreads and Liquidity" — which plots the Baa-Treasury corporate bond yield spread against a Treasury liquidity measure (the on-the-run/off-the-run yield spread) and reports a correlation of 0.96. Extend the series to the longest sample your data allow, and assess whether your results are consistent with the slide.

This project is primarily a test of whether your group can correctly apply the tools covered in class: constructing yield-spread series, cleaning and aligning data from several sources, computing and interpreting correlations, separating the default-risk and liquidity components of credit spreads using the structural (Merton) model and the credit spread puzzle, and testing how sensitive your results are to your assumptions. Each of these is listed below and graded.

Requirements

1. **State an economic question and your intended audience.** Although this is a replication exercise, frame it around a question the extended figure helps answer (e.g., how much of the corporate credit spread reflects liquidity rather than default risk? Does the relationship hold outside the 2007–2009 crisis?) and identify who would care about the answer (e.g., central bank policymakers designing liquidity facilities, or corporate bond fund managers).
2. Source or construct a Baa-Treasury yield spread (e.g., a Baa corporate bond yield minus a Treasury yield of comparable maturity) and a Treasury liquidity proxy such as the on-the-run/off-the-run yield spread, using CRSP, Compustat, and/or another reliable source (e.g., FRED). Document your source and construction for each series.
3. **Construct the series correctly.** Match maturities between the corporate and Treasury yields, express both series in consistent units (percent or basis points), and state how you convert to a common frequency (e.g., monthly averages vs. end-of-month values).
4. **Clean the data using the methods covered in class.** At a minimum, address missing values (e.g., holidays and blank entries in FRED series), duplicate observations, and outliers, and align series from different sources to a common set of dates. Winsorize (state the cut-offs) or explain why winsorizing is not appropriate — for example, if crisis-period spikes are the economic signal you are studying. Report how many observations each step affects.
5. Plot both series on a dual-axis chart over the original sample period shown on the slide (roughly Jan 2006 to late 2009) and compute the correlation between the two series.
6. Extend your plot and correlation estimate to the longest sample your data allow.
7. Compare your reproduced correlation and figure to the slide (Corr = 0.96). If they differ, explain why (e.g., differences in maturity matching, index construction, definition of the liquidity proxy, or sample period).
8. **Liquidity vs. default risk.** The structural (Merton) model covered in class implies that credit spreads widen when firms' asset volatility rises (slides 48–51). Using VIX as a proxy for default risk, regress the Baa-Treasury spread on your liquidity proxy and on VIX, separately and together, and report the coefficients, t-statistics, and R^2. Compare your VIX correlation with

slide 51 (Corr = 0.64), and discuss which factor explains more of the spread and how your results relate to the credit spread puzzle (slide 54). Note that VIX starts in 1990, so state the sample you use.

9. **Credit spread puzzle.** Following slides 55–58, estimate the expected default loss on Baa bonds (default probability × (1 – recovery rate)), using default and recovery rates from class or from Moody's, and compare it with the average Baa-Treasury spread in your sample. How much of the spread is left over after expected default losses, and how much of that residual can your liquidity evidence account for?
10. **Test how sensitive your results are to your assumptions.** In addition to the sample extension above, re-compute the correlation under at least two alternative assumptions, such as:
    - sub-periods (e.g., crisis vs. non-crisis years),
    - levels vs. changes in the two series,
    - a different Treasury maturity or a different liquidity proxy, or
    - a different data frequency (e.g., daily vs. monthly).

    Report whether the strong relationship on the slide holds up and whether your answer to the economic question changes. A compact table is sufficient.

Deliverables

- **Uploaded to the LEARN Dropbox 2 days before the presentation date:**
  - Code (script or notebook) that reproduces your results end-to-end, plus a README (see Reproducibility above).
  - The raw data files you actually used, trimmed to just what's needed to run your code (not the full downloaded dataset), so your submission runs end-to-end without extra downloads.
  - A report (max. 3 pages) containing:
    - Economic question and intended audience.
    - Data description: sources, construction of each series (maturity matching, units, frequency), and cleaning steps.
    - Your reproduced figure (extended sample) and correlation estimate.
    - A comparison to the slide 59 figure and correlation, with explanations for any differences.
    - Your liquidity vs. default risk regressions (one table).
    - Your credit spread puzzle decomposition.
    - Sensitivity analysis: how the correlation changes under at least two alternative assumptions.
    - Conclusion: what your results mean for your economic question and intended audience.
- **Uploaded to the LEARN Dropbox for slides 1 hour before the presentation session:** your presentation slides.
- **In class:** a 5–6 minute presentation of your own project, with every group member present to answer the instructor's questions, and at least one prepared question or comment during your paired group's presentation.

Grading Rubric (15 pts)

Criterion	Points
Economic question and intended audience clearly stated and motivated	1
Data sourcing, series construction (maturity matching, units, frequency), and cleaning correctly done and documented	2
Figure (dual-axis time series) and correlation correctly reproduced over the original and the longest available sample	2
Comparison to the original figure is thoughtful and differences are well explained	1
Liquidity vs. default risk regressions correctly specified, reported, and tied to the structural model	2
Credit spread puzzle decomposition correctly computed and interpreted	1
Sensitivity analysis: at least two alternative assumptions tested and their effect on the conclusion discussed	2
Reproducibility: submitted code and data reproduce the reported results without modification	1
In-class presentation: 5–6 min, clearly explaining your question, methodology, and results; each member present and answering the instructor's questions (graded individually; absent members receive a lower grade)	2
Peer review: substantive question or comment on the paired group's project during their presentation	1

Note: For all projects, keep your data pull, cleaning steps, and analysis reproducible from the raw source data. Cite any external packages or data sources you use. Every student must present at least once across the 6 project presentation sessions during the term.